

Ganesh Gupta

817-822-1131 | ganeshg11ta@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Kept large-scale **VM clusters** serving through **node loss and provider failures** by building **compute infrastructure** for **VM scheduling, fault-tolerant placement, and orchestration** in **Go, C++, and Python** across **AWS and GCP**.
- Owned the compute stack end to end, from **VM provisioning** and **cluster orchestration** to **resource allocation**, for long-running distributed training and high-availability inference on **NVIDIA GPU** fleets.
- Sustained **sub-200ms p99 latency** by building a **distributed executor** that routes workloads to **containers** or **serverless compute** via typed **REST APIs**, backed by an **S3** artifact store.
- Blocked failed releases before production by building **deployment gates** with **regression checks**, owning **design reviews** and **production debugging**.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** by rebuilding a synchronous service into an **async, autoscaled** architecture on **GCP**.
- Improved serving **performance** by extending inference-framework internals at the **kernel level (CUDA/Triton)**, including a **distributed KV cache**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Increased throughput **2.4x** and cut **memory usage 27%** by **profiling** compute and memory bottlenecks across **C++/Python** services and optimizing batching, scheduling, and **concurrency**.
- Operated **25+ production services** for **1M+ users** at **99.9% uptime** over 4 years on **AWS (EC2, RDS, SQS)**, owning **system design** and **operations**.
- Cut **production escalations 60%** in one quarter by standardizing **retry, timeout, and error-propagation** contracts across **distributed services**, driving **root cause analysis**.
- Cut production regressions **28%** and validation time from **6 hours to 90 minutes** by building an automated **deployment-gating** framework with **regression checks** and rollback criteria.

PROJECTS

VPC Control Plane Simulator

github.com/ganeshhgupta/vpc-control-plane-sim

- Built an AWS-style **cloud networking control plane** with **CIDR validation** and overlap detection, **longest-prefix-match routing**, stateful **security group** and stateless **NACL** evaluation, and configuration **drift detection**.

Distributed KV Store

github.com/ganeshhgupta/Distributed-KV-Store

- Built a replicated key-value store in **Go** implementing **Raft consensus**, **linearizable reads**, and **log compaction**, tolerating node failures without data loss.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Built a full-stack **system health** observability platform with **latency tracking**, **regression detection**, and alerting across distributed deployments, on **FastAPI** and **PostgreSQL**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: C++, Go, Python, Java, SQL

Compute Infrastructure: VM Scheduling, Fault-Tolerant Placement, Cluster Orchestration, Distributed Systems, Cloud Networking, GCP, AWS

Performance: Profiling, Concurrency, Memory Optimization, Production Debugging, Root Cause Analysis

Fundamentals: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems

Certifications: GCP Professional MLOps Engineer, AWS Certified Developer Associate